ExpV2S: Zero-Shot Expressive Video-to-Speech Synthesis via Latent Diffusion Model

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Abstract

Video-to-Speech (V2S) Synthesis aims to generate speech that synchronizes with lip motions and reflects the speaker's unique characteristics. However, existing methods tend to overlook the speaker's emotional state, which is crucial for producing expressive speech, and they often struggle to retain speaker's characteristics for unseen individuals, reducing the model's generalization capability. To address these issues, we propose ExpV2S: a latent diffusion model (LDM) for zero-shot **Exp**ressive **V**ideo**2S**peech synthesis. In general, we reformulate video-to-speech synthesis as a denoising process in a latent space, guided jointly by lip movements and facial features. More specifically, we introduce an emotion disentangling module to decompose the lip motions into seperate emotion and context embeddings by generating cross-reconstructed emotional speech. Then, a face-speech contrastive learning mechanism (FSCL) is proposed to capture the correspondence between visual and acoustic speaker characteristics, enabling accurate voice synthesis from a single face image. As a result, the proposed ExpV2S is capable of generating expressive speech that is both intelligible and emotionally consistent, and more importantly, it generalizes across different speakers. Extensive experiments demonstrate that our approach outperforms state-of-the-art methods in both quantitative metrics and human evaluations, effectively synthesizing expressive speech, especially for generalized zero-shot individuals. For more samples, please refer to the supplementary material.

CCS Concepts

• **Computing methodologies → Scene understanding**.

Keywords

Video-to-Speech Synthesis, Latent Diffusion Model, Emotion Disentangling, Zero-shot Generalization

ACM Reference Format:
Fang Zhang, Anqi Gou, and Linli Xu. 2026. ExpV2S: Zero-Shot Expressive Video-to-Speech Synthesis via Latent Diffusion Model. In *International Conference on Multimedia Retrieval (ICMR '26), June 16–19, 2026, Amsterdam, Netherlands.* ACM, New York, NY, USA, 10 pages. https://doi.org/10.1145/3805622.3810884

ACM ISBN 979-8-4007-2617-0/2026/06
https://doi.org/10.1145/3805622.3810884

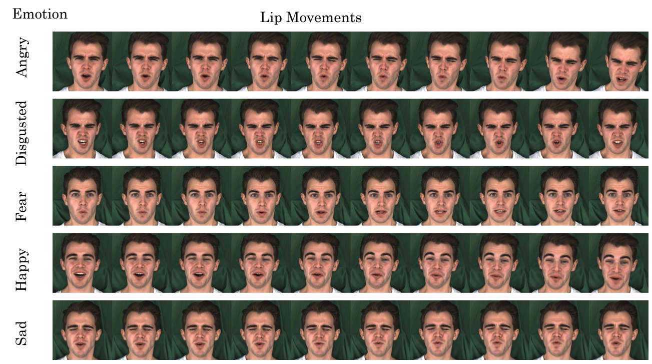

Figure 1: Visualization of lip movements which vary significantly when the same sentence is spoken in different emotional states.

1 Introduction

Video-to-Speech (V2S) synthesis is the task of generating the corresponding speech given a silent video of a talking person. This task has garnered growing interest due to its wide range of applications, including recovering speech in a noisy environment, providing reverse assistance for the hearing impaired and supporting linguistics research.

Recently, substantial efforts have been devoted to this research area to improve the quality of generated speech [11, 32], which primarily focus on speech intelligibility by leveraging the correspondence between the linguistic content in the speech and lip motions. For example, Mira et al. integrated both local and global visual features into GAN to establish a mapping function from visemes to phonemes, while Yemini et al. incorporated the text modality through an ASR model as classifier guidance during the diffusion process to alleviate lip motion ambiguity. However, emotion, a critical component of human speech, has yet never been considered in previous works. Intuitively, emotional state in facial expressions fundamentally alters speech articulation patterns, even for identical linguistic content. As illustrated in Figure 1, a speaker exhibits distinctly different lip motions across emotional states when saying the same sentence. For instance, when the speaker is angry, the lips tend to open wider and move rapidly, whereas in sadness, the lips are more likely to contract and move slowly. As lip motions are not only influenced by the content but also the emotions in the speech, it is challenging to explicitly exploit the complex relationships between lip motions and emotional speech.

A parallel line of research focuses on improving the generalization capabilities, so that even for unseen speakers, the V2S models would preserve the voice characteristics like gender, age and accent. To achieve that, several methods [7, 17, 28] extract voice-based speaker embeddings (VSEs) from the reference speech samples which are then utilized as conditions for generation. However, the reference speech may not always be available during inference, limiting their practical applicability. To address this limitation, several approaches have attempted to derive face-based speaker embeddings (FSEs) from face images in lieu of the reference speech. Sheng et al. transferred the knowledge from the speech identity encoder to the face identity encoder, enabling voice characteristic extraction from facial features through contrastive learning. DiffV2S [6] employed prompt tuning on the pretrained audio-visual model [38] to extract vision-guided speaker embedding representations. However, these methods still fall short in training a stable and generalized mapping from FSEs to VSEs. This can be attributed to the following challenges: i) The modality gap between acoustic and visual modalities complicates this mapping. ii) The limited number of speakers in most V2S datasets constrains the face encoder's ability to learn generalizable representations for unseen speakers.

To resolve the aforementioned issues, we propose ExpV2S: a latent diffusion model (LDM) for zero-shot expressive video-to-speech synthesis. First, we reformulate speech synthesis as a conditional denoising process in a latent space conditioned on facial appearan and lip movements. To take the speaker's emotions into consideration, instead of directly modeling the lip-to-speech translation, we decompose the complex lip motions through an emotion disentangling module that extracts distinct emotion and content embeddings. These two embeddings respectively provide supervision for generating spoken words and emotional state in the speech. Second, we bridge the gap between acoustic and visual modalities by encoding face images and the corresponding mel-spectrams into a unified space where the relationships between them can be more effectively exploited. Then face-speech contrastive learning is introduced to encourage the semantic alignment between FSEs and VSEs. By utilizing FSEs that closely align with VSEs in the latent space as conditions, our model demonstrates zero-shot generalization capabilities across different individuals.

To evaluate the performance of ExpV2S, in addition to the traditional V2S benchmarks such as LRS2 [40] and LRS3 [1], we also introduce an emotional talking head synthesis dataset: MEAD [44] to evaluate emotional synthesis capabilities and an open domain dataset to asses model's generalization capability. Comprehensive experiments demonstrate ExpV2S outperforms state-of-the-art V2S methods in both quantitative metrics and human evaluations. Our contributions are summarized as follows:

- We propose ExpV2S based on Latent Diffusion Model for expressive speech generation. With an emotion disentangling module, ExpV2S not only synchronizes with lip motions but also aligns with speakers' emotions.
- To enhance generalization to unseen people, we introduce a face-speech contrastive learning mechanism (FSCL) to enable accurate inference of voice characteristics from facial features alone, allowing high-quality speech synthesis for unseen speakers without reference speech.
- We conducted extensive experiments on multiple benchmarks, with quantitative and subjective evaluations, indicating that ExpV2S significantly outperforms the previous SOTA methods with strong zero-shot capablities.

2 Related Works

2.1 Video-to-Speech Synthesis

Previously, numerous studies have been conducted on video-to-speech synthesis due to its immense potential in the field of lip reading. To the best of our knowledge, Le Cornu and Milner were the first to propose a deep learning based network to reconstruct an audio clip by estimating a spectral envelope from visual features. Further, to enhance generalization for both seen and unseen speakers, several approaches [28, 43] extracted the voice-based speaker embeddings (VSEs) from the reference audio to control voice characteristics. For instance, SVTS [28] utilized a pre-trained speaker encoder to extract VSEs which are then combined with visual features as inputs of mel-spectrams generator. However, the extra reference audio is not always available during inference. Consequently, researchers have focused on producing the face-based speaker embeddings (FSEs) from solely visual information that reflects vocal characteristic. Lipvoicer[45] simply employed the face features extracted from Resnet-18 [12] as the condition for diffusion-based mel-spectrams decoder. Additonally, Sheng et al. introduced an associated voice-face representation learning strategy that facilitates the knowledge transfer from a speech identity encoder to the face identity encoder. This approach enables face identity encoder to extract necessary voice-based characteristics. Furthermore, DiffV2S [6] adopted prompt tuning method on the large pre-trained audio-visual model [38] to derive a visual-based speaker representation from video input.

2.2 Latent Diffusion Model

Diffusion Probabilistic Models (DPMs) [39] have demonstrated significant capabilities across various domains, including image generation [14, 38, 41], video generation [13, 20, 26] and audio synthesis [15, 21, 36]. A pivotal advancement in the evolution of DPMs is the proposal of Latent Diffusion Model (LDM) [34], which performs diffusion and denoising procedure in the latent space of powerful pretrained autoencoders. This approach substantially reduces computational cost compared with previous pixel space-based training and allows for robust and flexible conditioning inputs based on cross-attention mechanisms. In the filed of audio generation, researchers have begun to leverage the strengths of DPMs to enhance the quality of generated audio. Diffwave [18] is introduced for conditional and unconditional waveform generation based on a versatile diffusion probabilistic model. Several text-to-audio (TTA) systems [21, 22] have also employed latent diffusion models to achieve advantages in both quality and computational efficiency. To the best of our knowledge, ExpV2S is the first V2S approach to generate the mel-spectrams in the latent space.

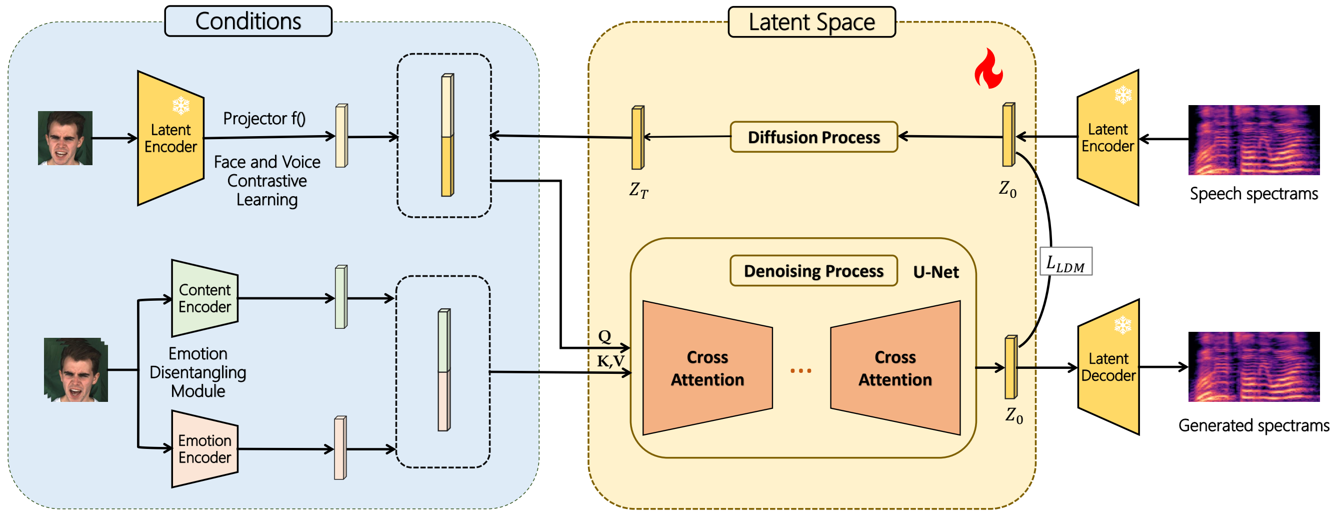

Figure 2: Overview of the proposed ExpV2S. Our framework is built on the latent diffusion model with the face-based speaker embedding, emotion embedding and content embedding as conditions. ExpV2S not only synthesizes expressive human speech synchronized with the lip motion but also conveys the speaker's emotions across different individuals.

3 Method

3.1 Overview

We present ExpV2S, a novel approach that reformulates expressive speech synthesis as a conditional denoising process in latent space, guided by both lip movements and facial features. An overview of ExpV2S is shown in Figure 2. We firstly train the projection layer with frozen latent encoder by Face-Speech Contrastive Learning (FSCL) to align FSEs and VSEs in Section 3.3. This alignment enables generalized speaker characteristic modeling from a single facial image, where the aligned FSEs serves as a driving factor in LDM for generalized speaker characteristics modeling. Secondly, we introduce an Emotion Disentangling Module (EDM) to extract emotion and content embeddings from lip motions and facial expressions respectively in Section 3.4. Guided by the cross-reconstruction loss, the disentangled emotion and content embeddings facilitate the reconstruction of both linguistic content and emotional expression in the synthetic speech.

3.2 Conditional LDM For EV2S

The emergence of Latent Diffusion Models (LDMs) [14, 34] provides a straightforward and effective way for high-fidelity image synthesis. Building on their strengths, we adopt LDMs as the foundation of our method and further explore its potential in synthesizing mel-spectrams from video inputs. Given a video sequence $V \in \mathbb{R}^{L\times H\times W\times 3}$ where L, H and W represent the frame length, height and width respectively, our objective is to reconstruct the corresponding speech mel-spectrams $M \in \mathbb{R}^{K\times S}$, where K is the number of mel filters and S denotes the number of time frames. A pretrained latent encoder $\mathcal{E}$ is used to extract the low dimensional latent image embeddings $z = \mathcal{E}(M) \in \mathbb{R}^{C_1\times \frac{K}{r}\times \frac{S}{r}}$, where C_1 is the latent dimensionality and r is the compression rate. The LDM is utilized to fit the data distribution $p(z)$ by denoising in the latent space.

For the forward process, noise is progressively added to the original latent representation with a fixed schedule $\alpha_1, ..., \alpha_T$, where T is the total number of timesteps and $\bar{\alpha}_t = \prod_{i=1}^{t} \alpha_i$. This process ultimately transforms the latent representation into a standard Gaussian distribution z_0 as follows:

$$\begin{aligned} q(z_t|z_{t-1}) &= \mathcal{N}(z_t; \sqrt{\alpha_t}z_{t-1}, (1-\alpha_t)\mathbf{I}) \\ q(z_t|z_0) &= \mathcal{N}(z_t; \sqrt{\bar{\alpha}_t}z_0, (1-\bar{\alpha}_t)\mathbf{I}) \end{aligned} \tag{1}$$

In the denoising process, the neural backbone $\epsilon(\cdot, t)$ of the standard LDM is implemented as a time-conditional UNet [35] which gradually learns the reverse process of a Markov Chain of length T. The goal of the LDM is to mirror score matching by optimizing the denoising objective:

$$\mathcal{L}_{LDM} = \mathbb{E}_{z_0,t,\epsilon\in\mathcal{N}(0,1)} \|\epsilon - \epsilon_\theta(z_t, t)\|_2^2 \tag{2}$$

Conditioning Mechanisms. Given the driven video, ExpV2S should satisfy three requirements: synchronization with lip motion, emotional consistency, and alignment with the speaker's characteristics. These are respectively achieved by utilizing face-based speaker characteristics, content and emotion embeddings as conditions for the LDM.

To extract FSEs, we randomly select a reference image $R \in \mathbb{R}^{H\times W}$ in the video sequence containing the speaker's visual characteristics. R is mapped into the latent space through a frozen encoder $\mathcal{E}$ and a projection layer $f(\cdot)$, to obtain FSEs $s = f(\mathcal{E}(R))) \in \mathbb{R}^{C_1}$. Since the speaker's characteristics remain constant over time, we extend the height and width of FSEs to match the dimensionality of the latent variable z, yielding $\dot{s} \in \mathbb{R}^{C_1\times \frac{K}{r}\times \frac{S}{r}}$. During training, after the latent representation of the mel-spectrams z_0 is transformed into the standard Gaussian distribution z_T, the speaker embedding is concatenated channel-wise with z_T to form the query $\{z_T, \dot{s}\}$ in the cross-attention module of the denoising network ϵ_θ.

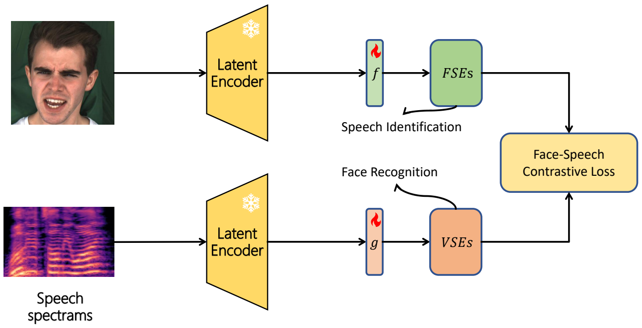

Figure 3: Overview of Face-Speech Contrastive Learning. A latent encoder projects both face images and the corresponding mel-spectrams into a unified space. Guided by a face-speech contrastive loss, FSEs are aligned and strongly correlated with VSEs.

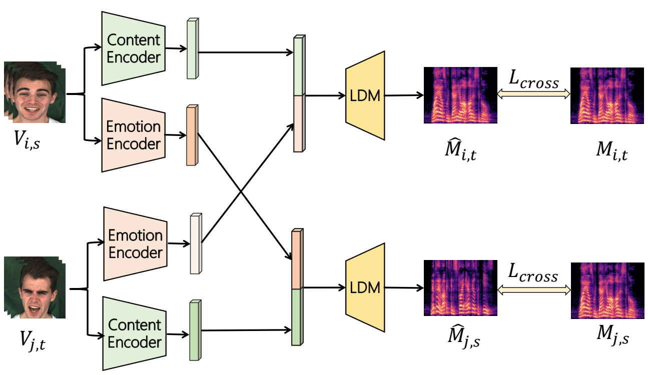

Figure 4: An illustration of the Emotion Disentangling Module. Various inputs of videos, conveying the same speaker but different emotions and contents, are processed to generate cross-reconstructed mel-spectrams representing different components of the speech.

For content embeddings, a content encoder E_c is utilized to extract the linguistic content embeddings from mouth regions of the video, yielding $c = E_c(V) \in \mathbb{R}^{L\times C_2}$ where L is the frame length and C_2 denotes the number of channels.

Similarly, we employ an emotion encoder E_e to extract emotion embeddings $e = E_e(V) \in \mathbb{R}^{L\times C_3}$ from facial expressions where C_3 represents the output dimensionality of E_e. By combining $\{s, e\}$ as the key and value in the cross-attention module, ExpV2S can be formulated as a conditional denoising process optimized with the following objective:

$$\mathcal{L}_{LDM} = \mathbb{E}_{z_0,t,\epsilon}\|\epsilon - \epsilon_\theta(z_t, t, s, e, c)\|_2^2 \tag{3}$$

3.3 Face-Speech Contrastive Learning

To control the voice characteristics of any individuals based on facial images, we employ the Face-Speech Contrastive Learning (FSCL) to align FSEs and VSEs within a unified latent space to bridge the modality gap. We observe that the latent encoder $\mathcal{E}$ of Stabale Diffusion (SD) can extract speaker-related features from both face image and mel-spectrams as mentioned in Section 6.1. This insight inspires us to align their representations within a unified space where the relationship between them is easier to exploit. Furthermore, $\mathcal{E}$ is pretrained on a large dataset LAION-5B, and thus provides comprehensive facial features for any individuals, thereby enhancing the model's zero-shot capability.

As illustrated in Figure 3, given a face image R and a mel-spectram M, we first utilize a frozen encoder $\mathcal{E}$ to extract rich semantic features for both of them. Two trainable projection layers $f(\cdot)$ and $g(\cdot)$ take these semantic features as inputs and are pre-trained on the speech identification and face recognition tasks, respectively. The objective of this pre-training stage is to derive face-based speaker embeddings $s = f(\mathcal{E}(R))$ and voice-based speaker embeddings $u = g((\mathcal{E}(M))$. Subsequently, the face-speech contrastive loss is employed to align FSEs and VSEs as follows:

$$\mathcal{L}_{FSCL} = -\frac{1}{N}\sum_{i=1}^{N}\log\frac{\exp(\text{sim}(s_i, u_i)/\tau)}{\sum_{j=1}^{K}\exp(\text{sim}(s_i, u_j)/\tau)} \tag{4}$$

where $sim(\cdot)$ is the cosine similarity, N is the number of batch size and (s_i, u_i) belongs to the positive pairs of face and speech samples. Notably, our goal is to maximize the similarity between face-speech pairs from the same speaker and minimize the similarity of face-speech pairs from different speakers. Thus, the positive pairs consist of samples belonging to the same speaker, rather than face-speech pairs from the same video. The FSCL mechanism jointly preserves speaker-specific characteristics while establishing robust semantic alignments between FSEs and VSEs across different speakers.

Following FSCL, the aligned FSEs can effectively capture the corresponding speaker's voice characteristics and subsequently serve as conditions, as discussed in Section 3.2.

3.4 Emotion Disentangling Module

As the lip motions are influenced simultaneously by spoken words and emotional states, learning the intricate relationship between speech and lip motions directly is highly challenging. To address this issue, we propose an Emotion Disentangling Module (EDM) that separates lip motions into distinct content and emotion embeddings, guided by cross-reconstruction training loss.

The overview of emotion disentangling training is shown in Figure 4. Two pre-trained encoders E_c and E_e are used as feature extractors, which are pre-trained on lip reading and emotion classification tasks, respectively. However, the encoders cannot guarantee the disentanglement between content and emotion features. To compel the encoders to acquire disentangled content and emotion representations, we utilize video sequences that combine diverse emotions and linguistic contents as inputs, and train the network to reconstruct the corresponding ground truth samples. Specifically, given two video sequences $V_{i,s}$ and $V_{i,t}$ where the subscripts indicate the person in the video speaks the i-th utterance with the j-th emotion, the content encoder focuses on mouth regions to capture linguistic information, while the emotion encoder processes full facial expressions to extract emotional attributes, as the spoken words and emotional states are more prominently reflected in the mouth region and facial expression respectively.

The content embedding $c_i = E_c(V_{i,s})$ and emotion embedding $e_t = E_e(V_{j,t})$ are concatenated together as conditions to generate the corresponding sample $\hat{M}_{i,t}$ with the i-th utterance and the s-th emotion. By changing the condition set in $\mathcal{L}_{LDM}$, we formulate the cross-reconstruction loss as follows:

$$\mathcal{L}_{cross} = \mathbb{E}_{z_0,t,\epsilon} \|\epsilon - \epsilon_\theta(z_t, t, s, E_c(V_{i,s}), E_e(V_{j,t})\|_2^2 \quad (5)$$

where z_0 is the latent embedding of ground-truth mel-spectrams $M_{i,t}$. This loss promotes disentanglement between emotion and content embeddings by ensuring that they can be recombined to accurately reproduce both spoken words and emotional expression present in the speech.

3.5 Loss Function

The overall loss function is a weighted sum of two distinct components: the cross-reconstruction loss and the latent diffusion loss. It is defined as:

$$\mathcal{L} = \mathcal{L}_{LDM} + \lambda_1 \mathcal{L}_{cross} \quad (6)$$

where λ_1 is the hyper-parameter. Note that $\mathcal{L}_{cross}$ is exclusively applied to emotional speech datasets, while for standard V2S datasets, optimization relies solely on the LDM reconstruction objective $\mathcal{L}_{LDM}$ for mel-spectrogram synthesis.

4 Experiments

4.1 Datasets

V2S Datasets. In line with previous methods, we conduct comparisons against other baselines on the traditional V2S datasets including LRS2-BBC[40] and LRS3-TED[1]datasets. LRS2-BBC, collected from BBC television shows, contains over 200 hours of videos, with over 143,000 utterances are divided into pre-train and train-val splits. We use both the pre-train and train-val datasets for training, reserving 1,100 utterances for inference[6]. LRS3-TED is derived from TED and TEDx videos, recording over 9000 different speakers with 150,000 utterances across approximately 440 hours. Following the approach outlined in [17], we evaluate our model on the complete unseen test data splits.

Emotional V2S Datasets. We have redesigned and restructured existing emotional talking head datasets, specifically MEAD [44] and EmoVOCA [30], to develop specialized emotional V2S datasets. We utilize these talking head resources to evaluate the expressiveness of the generated speech rather than to assess the quality of the produced virtual talking heads. Specifically, the MEAD dataset contains audio-visual recordings of the same sentence across varying emotional states, which satisfies the requirements for cross-reconstruction training. We select 34 actors for the training set and 6 actors (consisting of 3 male and 3 female unseen speakers) for the test set. EmoVOCA [30] is a large-scale synthetic 3D emotional talking head dataset that realistically entangles speech-related motions with expression dynamics. It combines inexpressive 3D talking heads from VOCAset [10] with emotional facial expressions from the Florence 4D dataset [33], providing a diverse set of 11 emotions across three intensity levels. In this work, we utilize the EmoVOCAv2 split, which contains 15,840 sequences. Following Nocentini et al., the data are divided into training (8 actors), validation (2 actors), and test sets (2 actors). These two talking head datasets are leveraged to assess whether the proposed model can accurately synthesize expressive speech synchronized with subtle lip movements.

4.2 Implementation Details

For the latent diffusion models, we follow the Diff-Foley [25] to encode single channel mel-spectrams by repeating the spectrams channel into ($m \rightarrow m \in \mathbb{R}^{3 \times K \times S}$). We use only the first channel from the decoded latent representation for decoding $D_i(z_0[0,:,:])$. LDM is initialized with the pretrained SD-V2.1 where the channel number is $c_1 = 4$ and the compression rate is $r = 8$. The architecture of the content encode is based on a pretrained lip-reading model [27]. For the emotion encoder, we replace the first convolutional layer in MoCo v2 [5] with a 3D convolution layer. We set 1000 timesteps (T = 1000) as default and utilize DDIM [41] for faster sampling. DiffWave [18] as a vocoder to transform mel-spectrams into audio.For training, we apply an AdamW[24] optimizer with hyperparameters $\beta_1 = 0.9$, $\beta_2 = 0.98$ to train LDM with $\mathcal{L}_{LDM}$ on the LRS3 and LRS2 datasets. Then, we finetune the LDM on MEAD dataset with the weighted sum of $\mathcal{L}_{LDM}$ and $\mathcal{L}_{cross}$.

4.3 Evaluation Metrics

For quantitative metrics, we measure the synchronization between the generated waveform and the matching video by utilizing SyncNet [8] to calculate the temporal distance (LSE-D) and the confidence scores (LSE-C). To measure the low-level quality of the speech, we utilize Extended STOI (ESTOI) [16] to measure the intelligibility of the generated speech. We utilize Word Error Rate (WER) in order to evaluate the content quality of the generated speech. To evaluate how much the generated speech and the original have similar speaker characteristics, we introduce the Speaker Encoder Cosine Similarity (SECS) [3] by calculating the similarity of speaker embedding extracted from the speaker encoder [42]. Further, we adopt an off-the-shelf emotion classification network and calculate the emotion accuracy between predicted emotion embeddings and ground truth labels.

For subjective metrics, we conduct a human subjective evaluation using mean opinion scores (MOS) to assess intelligibility, naturalness, synchronization, voice matching, and emotion matching. Intelligibility and naturalness measure the clarity of the generated words. Synchronization determines the alignment between the generated audio and the lip movements in the video. Voice matching assesses how well the synthetic speech preserves the speaker's unique vocal characteristics. Furthermore, emotion matching is introduced to evaluate whether the synthesized speech accurately reflects the emotional state of the speaker as portrayed in the visual expressions.

4.4 Baselines

We compare ExpV2S with recent video to speech synthesis baselines. DiffV2S[6] and LipVoicer[45] adopt diffusion models as video to mel-spectrams generator with face images as guidance. SVTS [28] and Lip2Speech [17] utilize the reference speech for extracting the speaker embedding, we reproduce both methods disregarding the audio guidance for fair comparison following DiffV2S.

Dataset Type	Method	ESTOI ↑	LSE-C ↑	LSE-D ↓	WER ↓	Emotion Accuracy ↑	SECS ↑
V2S Dataset (LRS2-BBC)	Lip2Speech	**0.322**	7.19	7.01	61.0%	-	0.457
	SVTS	<u>0.301</u>	**7.87**	**6.30**	76.6%	-	0.499
	LipVoicer	0.230	6.60	7.84	<u>17.8%</u>	-	0.573
	DiffV2S	0.283	7.51	6.90	52.7%	-	<u>0.581</u>
	ExpV2S	0.292	<u>7.68</u>	<u>6.57</u>	**16.2%**	-	**0.701**
V2S Dataset (LRS3-TED)	Lip2Speech	0.240	4.85	9.15	74.8%	-	0.495
	SVTS	0.244	7.08	<u>7.04</u>	81.9%	-	0.543
	LipVoicer	0.210	6.23	8.26	<u>21.4%</u>	-	-
	DiffV2S	**0.284**	<u>7.28</u>	7.27	39.2%	-	0.625
	ExpV2S	<u>0.267</u>	**8.01**	**6.79**	**18.3%**	-	**0.759**
Emotional V2S (MEAD)	Lip2Speech	**0.279**	4.12	<u>8.73</u>	79.4%	0.243	0.129
	LipVoicer	0.185	<u>6.54</u>	9.01	26.1%	<u>0.304</u>	<u>0.594</u>
	ExpV2S	<u>0.255</u>	**7.23**	**7.57**	**14.2%**	**0.845**	**0.809**
Emotional V2S (EmoVOCA)	Lip2Speech	**0.275**	3.27	9.25	82.3%	0.172	0.235
	LipVoicer	0.194	6.31	<u>8.74</u>	<u>41.3%</u>	<u>0.159</u>	<u>0.528</u>
	ExpV2S	<u>0.263</u>	**7.05**	**6.12**	**16.8%**	**0.631**	**0.712**

Table 1: Objective evaluation on standard V2S and emotional V2S datasets. The best performance is highlighted in bold (1st best) and <u>underline</u> (2nd best). Some results are reported by DiffV2S [6]. Note that the emotion accuracy is not reported for the LRS2 and LRS3 datasets as the emotional states of most samples are neutral

5 Results and Analysis

5.1 Quantitative Evaluation Results

Comparisons on LRS datasets. We conduct a comprehensive quantitative analysis comparing ExpV2S and other V2S baselines as depicted in Table 1. For the LRS2 and LRS3 datasets, ExpV2S achieves the best or second-best performance on LSE-C and LSE-D metrics, indicating that the generated speech is well synchronized with lip movements. We obtain a score of 0.292 on the ESTOI metric, which is slightly lower than that of Lip2Speech and SVTS. However, both of them struggle to effectively capture the content in the speech as evidenced by lower WER metric. In contrast, our model outperforms all other baselines in the WER scores, obtaining 16.2% and 20.3% for LRS2 and LRS3 datasets. This highlights that ExpV2S is better at capturing correlations between content embeddings and spoken words compared to previous model. It is remarkable that ExpV2S also surpasses other models on the SECS metric by a significant margin, highlighting its strong ability to retain speaker's characteristics.

Comparisons on datasets with emotions. We now move on evaluating MEAD and EomVOCA datasets that contains speech with different emotions. As is shown in Table 1, other models not only struggle to generate emotion-consistent speech accurate but also exhibits a significant decline in speech intelligibility as indicated by emotion accuracy and WER scores. This observation supports our hypothesis that the speaker's emotions influence lip movements, thereby complicating the establishment of a clear relationship between lip motion and speech. Given that Emotion Disentangling Module effectively decomposes lip motions into emotion and content embeddings, our model is able to learn two relatively straightforward mappings: one from linguistic content to spoken words and another from facial expression to emotions, which facilitate coherent control over emotional speech. For instance, ExpV2S maintains a low WER of 14.2% and a high emotion accuracy of 0.845 for MEAD datasets.

5.2 Qualitative Results

We visualize the mel-spectrogram samples generated by the prior works LipVoicer and Lip2Speech, as well as our proposed method, alongside the ground truth mel-spectrograms. Figure 5 illustrates the samples from the MEAD dataset with happy emotional states. It is evident that the samples generated by LipVoicer tend to be blurry and fail to present the intricate details found in the ground truth mel-spectrograms. This trend ultimately degrades the waveform generation, leading to acoustic noise. While the results from Lip2Speech appear to exhibit some mel-spectrogram details, they fail to fully reconstruct the actual spectrogram compared to the ground truth and do not align with the temporal trajectory of the lip movements. Consequently, this misalignment results in incorrect speech audio. In contrast, the mel-spectrogram samples generated by our proposed ExpV2S method not only capture fine-grained mel-frequency details but also demonstrate a high visual correlation with the ground truth. These accurate and detail-rich mel-spectrograms facilitate the generation of speech with correct linguistic content and emotion states.

5.3 Human Subjective Evaluation

We also conduct a human subjective study through mean opinion scores (MOS). ExpV2S outperforms all baseline methods across all five evaluation metrics as depicted in Table 2. In terms of intelligibility, ExpV2S achieves a MOS of 3.85, while for naturalness and synchronization, it scores 3.67 and 3.99, indicating that our model effectively preserves clear content and achieves high-fidelity alignment with lip movements.

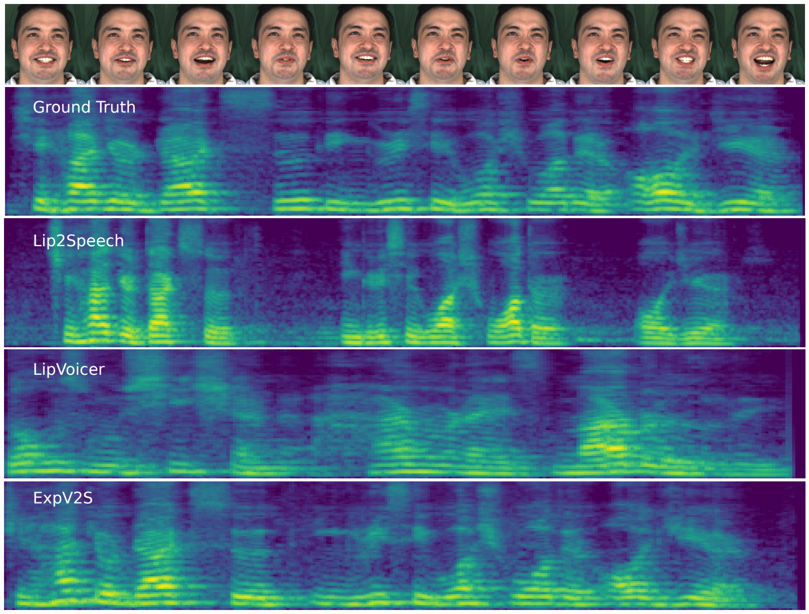

Figure 5: Visualization of generated mel spectrums with different methods on emotional V2S dataset. The top row denotes the facial motions in the video. It can be seen that the generated mel spectrums of ours align with the ground truth well.

Notably, our approach achieves a voice matching score of 4.03, which is comparable to the ground truth. This result confirms that ExpV2S can realistically reproduce the speaker's characteristics. Furthermore, ExpV2S demonstrates a significant advantage in emotion matching with a score of 3.91, far surpassing other V2S baselines. This indicates that the Emotion Disentangling Module (EDM) successfully captures and reflects the speaker's emotional state from facial expressions. Compared to TTS-based models like EmotiVoice, our model shows superior capability in modeling emotional nuances directly inferred from facial expressions.

5.4 Zero-shot Generalization Capability

To study model's generalization to unseen datasets, we construct an open domain dataset by randomly selecting 100 samples from audio-visual datasets, including CREAM [2], GRID [9] and RAVDESS [23], In addition to comparison with V2S models, we employ the state-of-the-art TTS model including GradTTS [31], EmotiVoice [1] and CloneVoice [2] to synthesize speech conditioned on the ground truth transcribed text. The results are shown in Table 3. Our approach outperforms all the V2S models by a considerable margin, demonstrating the strong zero-shot capability of ExpV2S. Compared with EmotiVoice, which synthesis emotional speech guided by ground truth emotion labels, ExpV2S outperforms it in emotion accuracy, demonstrating the strong capability of modeling emotion states inferred from facial expression. In addition, our model achieves highest SECS scores, while CloneVoice that utilizes a reference speech to extract speaker embeddings obtains a lower SECS. This demonstrates that ExpV2S synthesizes high quality speech even for unseen individuals.

[1] https://github.com/netease-youdao/EmotiVoice
[2] https://github.com/jianchang512/clone-voice

6 Analysis and Ablation Study

6.1 Face and Speech Contrastive Learning

Latent Encoder Analysis. To investigate the latent encoder's ability to extract generalizable and comprehensive speaker features, we follow the widely used linear probing [4, 46], where a linear classifier is trained on top of the frozen latent encoder to measure the top-1 accuracy on both speech identification and face recognition tasks. The results are shown in Table 5. Despite the latent encoder being frozen, we still achieved high accuracy scores of 84.3% for speech identification task and 90.5% for face recognition Task. These results indicate that the semantic features extracted from the latent encoder contain rich speaker characteristics. Therefore, training only a linear classifier can yield excellent results. In addition, we fine-tuned the entire model with 5% samples of the dataset, and it achieved comparable results even when using the full dataset. This suggests that the latent encoder inherently extracts speaker-related features, rather than learning them through supervised learning. These results further validate that the premise of projecting both face images and mel-spectrams into a unified latent space of $\mathcal{E}$ is reasonable.

Ablation Study of FSCL. To validate the efficiency of our face-speech contrastive learning, we conducted ablation study on LRS3 dataset to analyze how our FSCL contribute to preserving speaker's characteristics.

As is shown in Table 4, we remove $\mathcal{L}_{FSCL}$ prevent the alignment between FSEs and VSEs. Directly leveraging unaligned FSEs leads to a performance drop due to the visual-acoustic modality gap. Furthermore, we replace the aligned FSEs with the ground truth VSEs as the input of conditional set, this substitution yields only a minimal reduction in SECS, demonstrating that FSCL significantly encourages the semantic alignment between FSEs and VSEs. Besides, we replace FSEs with the speaker embeddings extracted from ground

Method	Intelligibility ↑	Naturalness ↑	Synchronization ↑	Voice Matching ↑	Emotion Matching ↑
Ground Truth	4.33	4.45	4.52	4.37	4.41
Lip2Speech	2.17	1.85	1.81	2.44	2.05
LipVoicer	2.52	2.11	2.89	1.95	1.18
GradTTS	3.65	3.58	3.01	3.45	1.23
EmotiVoice	3.44	3.47	3.29	3.10	3.02
ExpV2S (Ours)	**3.85**	**3.67**	**3.99**	**4.03**	**3.91**

Table 2: Human evaluation (Mean Opinion Score, MOS) on the MEAD dataset. Scores range from 1 to 5, where higher values indicate better performance.

Model	WER	Emotion Accuracy	SECS
V2S models			
LipVoicer	34.2%	0.169	0.423
Lip2Speech	78.6%	0.127	0.372
ExpV2S	24.5%	0.756	0.724
TTS-based models			
CloneVoice	2.2%	0.08	0.683
EmotiVoice	1.8%	0.698	0.201

Table 3: Zero-shot performance on the open domain dataset.

Settings	SECS
ExpV2S	0.759
w/o $\mathcal{L}_{FSCL}$	0.537
replace with VSEs	0.796
speaker embedding from speech identity encoder	0.498
replace positive pairs	0.325

Table 4: Ablation study of different components in FSCL mechanism. The SECS scores are evaluated on the LRS3

Setting	Speech Identification	Face recognition
linear probing	0.843	0.905
5% dataset	0.821	0.835
100% dataset	0.858	0.927

Table 5: The top-1 accuracy under speech identification and face recognition tasks evaluated on LRS3 dataset.

truth speech by a speaker verification encoder [42]. The SECS scores are lower than that of VSEs extracted from the latent encoder. This result is reasonable due to the advantage that $\mathcal{E}$ is pretrained on the large dataset while the speaker encoder is only trained on a small limited speakers. In terms of the loss function, replacing the positive samples with face-speech pairs from the same video resulting a significant drop. We assume that these samples that belongs to same speakers but vary in spoken words and emotions, may be split into the negative pairs. Thus, other factors such as emotions and linguistic contents prevent FSCL to focusing on extraction of speaker-related features.

6.2 Ablation Study of Emotion Disentangling Module

To verify the effectiveness of our Emotion Disentangling Module (EDM), we conducted an ablation study comparing the full EDM

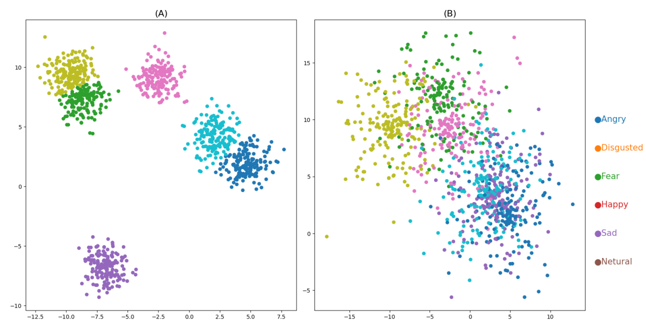

Figure 6: t-SNE visualization of the emotion encoder embeddings trained with (A) and without (B) the cross-reconstruction loss $\mathcal{L}_{cross}$. Each point stands for one utterance. Points with same Colors indicate the same content.

against a variant where the cross-reconstruction loss $\mathcal{L}_{cross}$ was removed. We visualized the latent space of the emotion encoder using t-SNE for both settings as illustrated in Figure 6. In this visualization, where points of the same color represent different sentences spoken with identical emotions, it is evident that samples sharing emotional semantics are more effectively clustered when the cross-reconstruction loss $\mathcal{L}_{cross}$ is utilized.

This observation indicates that our EDM strategy can effectively extract corresponding emotion embeddings from facial expressions, whereas the standard image encoder alone fails to capture such fine-grained emotional information. These results demonstrate that the cross-reconstruction mechanism is crucial for decoupling emotional details from complex facial motions, thereby ensuring the expressiveness of the synthesized speech.

7 Conclusion

We propose ExpV2S:a latent diffusion model (LDM) for zero-shot expressive Video2Speech synthesis. To enhance the expressiveness in the synthetic speech, we firstly take the speaker's emotion into the consideration. By decomposing the lip movements into emotion-related and content-related features with an Emotion Disentangling Module (EDM), ExpV2S is enabled to generatespeech that not only synchronizes with lip motions, but also retains the speaker's emotion. Furthermore, Face-Speech Contrastive Learning (FSCL) is introduced to empower the zero-shot speaker adaptation by aligning the face-based speaker embeddings with voice-based speaker embeddings in an unified latent space. Extensive experiments demonstrate that ExpV2S outperforms other baselines in quantitative metrics and human evaluations.

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